SOFTWARE TESTING AND QUALITY ASSURANCE

DEFECT ANALYSIS REPORT

Defect Register, Test-Failure Analysis and Priority–Severity Matrix

AlumniConnect — Alumni Networking and Mentorship Platform

Prepared as an evidence-based project testing and defect-tracking document.

Project	AlumniConnect
Report type	Defect analysis, test-case results and triage matrix
Execution date	28 October 2026
Data basis	Five existing project test scripts; 32 reported test cases
Build/version	Not identified by the executed test scripts
Prepared by	Not specified for this execution record

This report distinguishes test failures and verified implementation mismatches from confirmed user-reported production defects. It does not claim that unexecuted checks passed.

1. Defect Analysis Overview

AlumniConnect is a web platform for institutional alumni networking, mentorship, profiles and related community workflows. This report follows the defect-analysis structure of the supplied reference while using evidence from the project's existing tests.

Five local Node.js scripts were executed: font/typography checks, sidebar/layout checks, demo-account and helper checks, alumni import/privacy checks, and duplicate-registration scenarios. They reported **28 passes and 4 failures across 32 cases**. The four failures concern font configuration assertions.

> **Evidence boundary:** The typography scripts primarily inspect source text and configuration. Their failures demonstrate mismatches with the tests' stated expectations; they do not, on their own, prove a production incident. The report therefore records two requirements-conformance findings and one exact-string test discrepancy for owner triage—not a user-reported defect.

Scripts that reseed databases or contact live infrastructure were deliberately excluded. No production data was changed, and no new test cases were invented for this report.

2. Defect Summary by Severity

Severity	Definition for this project	Confirmed from executed tests	Interpretation
Critical	Broad outage, serious data loss, or confirmed severe security exposure.	0	No such defect was demonstrated by the selected tests.
High	Core workflow blocked or major privacy/access-control failure demonstrated.	0	No high-severity behavior failure was demonstrated in this run.
Medium	Visible functional or requirement-conformance issue with bounded impact.	2 findings, pending owner acceptance of the font requirement	Missing Helvetica choice and configured Roboto despite the checked-in tests' strict approved-font expectations.
Low	Minor formatting, fallback, or test-contract discrepancy.	1 discrepancy	Exact Times New Roman stack assertion differs because the configured stack includes an additional Georgia fallback.
Total	Findings from the selected test run, not a full release-cycle inventory.	3 findings; 4 failed assertions	Two tests report the missing Helvetica choice; the other two assertions identify the fallback and Roboto mismatches.

Severity is provisional. The project owner should confirm whether the existing font assertions represent the approved acceptance criteria before final priority assignment.

3. Defect Summary by Module

Module / workflow	Failed assertions	Potential impact	Recommended action
Typography / font selection	3	Available font choices and active typography may not match the project's stated font policy.	Confirm approved fonts; reconcile implementation and tests; rerun both typography scripts.

Module / workflow	Failed assertions	Potential impact	Recommended action
Sidebar / responsive layout	1	The font-choice assertion in the UI suite fails. Other sidebar and breakpoint source assertions passed.	Retest the font option after the requirement is resolved; treat breakpoint checks as source-only evidence.
Authentication / demo-account helpers	0	No failure in the selected fixture/helper assertions.	Do not infer full API authentication coverage from these local checks.
Alumni import / privacy sanitization	0	No failure in the tested dataset, normalization and sanitizer assertions.	Database persistence and broader authorization remain outside this run.
Duplicate registration	0	Local normalization and simulated response/data-preservation assertions passed.	Live controller and database behavior were not exercised.

4. Defect Register — Findings from the Executed Tests

QA-TYP-01 — Expected Helvetica option is absent

Field	Record
Type / status	Requirements-conformance finding; observed by existing automated assertions; owner triage pending.
Evidence	`testFontSelectorAndTypography.js` case 1 and `testSidebarControlsAndResponsiveness.js` case 5 both failed because the checked source did not contain the expected Helvetica option.
Expected	Font configuration provides the approved Arial, Helvetica, and Times New Roman choices as defined by those existing tests.
Observed	The test scripts report “Helvetica must be defined” / “Must include Helvetica”.
Impact / provisional severity	Font-selection requirement may be unmet; Medium, subject to owner confirming the requirement is current.
Priority suggestion	P2 if the approved typography requirement is mandatory for the current submission; otherwise re-triage against the current design decision.
Fix status	Not changed in this report task; no fix has been claimed.

QA-TYP-02 — Roboto is configured despite the test's restricted-font expectation

Field	Record
Type / status	Potential typography policy deviation observed by a repository test; owner triage pending.
Evidence	`testFontSelectorAndTypography.js` case 8 failed and reported “Disallowed web font found: roboto”. The inspected theme/CSS sources include a Roboto option/reference.
Expected	The test expects zero unauthorized external web-font names in inspected typography sources.
Observed	Roboto is present in the inspected source set, so the existing assertion fails.
Impact / provisional severity	Potential violation of the restricted font policy; Medium if Arial/Helvetica/Times-only remains binding.
Priority suggestion	P2 if the restricted-font requirement is confirmed; otherwise update the test contract to match the approved design.
Fix status	Not changed in this report task; no fix has been claimed.

QA-TYP-03 — Exact Times New Roman fallback string differs

Field	Record
Type / status	Test expectation discrepancy; functional font failure not established.
Evidence	`testFontSelectorAndTypography.js` case 3 failed because the actual configured stack contains an additional Georgia fallback compared with the exact string expected by the test.
Expected	The test requires an exact Times New Roman stack string.
Observed	Times New Roman is present, but the stack differs in its fallback list.
Impact / provisional severity	Low pending a decision on whether the extra fallback violates the requirement.
Priority suggestion	P3; align the test and implementation with the approved font-stack requirement.
Fix status	Not changed in this report task.

5. Test Cases — Existing Project Scripts Only

The cases below correspond only to test blocks/assertions in the five executed project scripts. “PASS” means the script's stated assertion passed; it does not mean a full end-to-end feature flow was tested.

5.1 Font Selector and Typography — `testFontSelectorAndTypography.js`

ID	Test / expected result	Observed result	Status
TC-TYP-01	Theme source maps the approved Arial, Helvetica and Times New Roman font definitions.	Helvetica definition not found by the test.	FAIL
TC-TYP-02	Font application sets `--app-font-family`, persists the choice and dispatches the font-change event.	All checked source markers present.	PASS
TC-TYP-03	Pre-hydration script restores the expected exact font stacks before first paint.	Times stack differs from the exact string expected; Georgia fallback is present.	FAIL
TC-TYP-04	Global CSS binds page typography and form controls to the active font variable.	All checked CSS markers present.	PASS
TC-TYP-05	Tailwind sans and display families bind to the active font variable.	Both expected bindings present.	PASS
TC-TYP-06	Application startup applies the initial/persisted font and theme.	Both startup calls found in source.	PASS
TC-TYP-07	Toolbar listens to font/theme/storage changes and exposes the expected accessible label.	Checked event and label markers present.	PASS
TC-TYP-08	Inspected typography sources contain no disallowed external web-font names.	Roboto reference detected by the assertion.	FAIL

5.2 Sidebar, Theme Controls and Breakpoint Source Checks — `testSidebarControlsAndResponsiveness.js`

ID	Test / expected result	Observed result	Status
TC-UI-01	Desktop sidebar source places controls above the profile card with expected layout markers.	Expected source markers and ordering present.	PASS
TC-UI-02	Font selector and theme toggle use the expected horizontal row sizing.	Expected class strings present.	PASS
TC-UI-03	Font selector source includes the type icon, label and dropdown indicator.	Expected component markers present.	PASS
TC-UI-04	Theme toggle source uses the expected Moon/Sun icons and action labels.	Expected markers present.	PASS
TC-UI-05	Theme source provides Arial, Helvetica and Times New Roman with expected stacks.	Helvetica option not found.	FAIL
TC-UI-06	Dropdown source supports Escape/arrow keys and listbox ARIA semantics.	Expected keyboard and ARIA markers present.	PASS
TC-UI-07	Mobile drawer source includes the toolbar and profile section.	Expected source markers present.	PASS

ID	Test / expected result	Observed result	Status
TC-UI-08	Source includes mobile/desktop layout class patterns for 320, 375, 768, 1024, 1280 and 1440 px scenarios.	Expected class markers present; no browser rendering was performed.	**PASS**

5.3 Demo Accounts, Password Helpers, Email Rules and URL/Privacy Helpers — `testAuthAndDemoAccounts.js`

ID	Test / expected result	Observed result	Status
TC-AUTH-01	Configured admin demo account has the expected role, institutional email and active/verified state.	Assertions passed.	**PASS**
TC-AUTH-02	Configured teacher demo account has the expected role, institutional email and active/verified state.	Assertions passed.	**PASS**
TC-AUTH-03	Configured student demo account has the expected role, institutional email and active/verified state.	Assertions passed.	**PASS**
TC-AUTH-04	Configured alumni demo account has the expected role, institutional email and active/verified state.	Assertions passed.	**PASS**
TC-AUTH-05	Bcrypt accepts the matching password and rejects an incorrect password.	Both comparisons passed.	**PASS**
TC-AUTH-06	Local email-rule examples accept the institutional domain and reject public/other domains.	All included examples passed; registration API not called.	**PASS**
TC-AUTH-07	LinkedIn helper normalizes accepted URLs and rejects malformed/javascript-scheme inputs.	All included helper assertions passed.	**PASS**
TC-AUTH-08	Public profile sanitizer strips phone fields and hides private/unconfirmed LinkedIn data.	All included sanitizer assertions passed.	**PASS**

5.4 2022 Alumni Data and Privacy — `test2022ImportAndPrivacy.js`

ID	Test / expected result	Observed result	Status
TC-DATA-01	Import dataset contains 48 entries and no phone/secondary-phone fields.	All 48 records passed the asserted field checks.	**PASS**
TC-DATA-02	Company normalization handles known variants and empty/null values.	All included normalization assertions passed.	**PASS**
TC-DATA-03	Sanitizer enforces LinkedIn visibility for public, student, permitted and admin viewer fixtures while stripping phone fields.	All four viewer scenarios passed.	**PASS**

5.5 Duplicate Registration Scenarios — `testDuplicateRegistrationFlow.js`

ID	Test / expected result	Observed result	Status
TC-DUP-01	Mixed-case/whitespace email examples normalize consistently.	All examples passed.	**PASS**
TC-DUP-02	Local validation examples reject external domains and malformed email values.	All examples passed; no API call was made.	**PASS**
TC-DUP-03	Simulated duplicate response has expected 409 status/message and omits checked sensitive fields.	In-script response simulation matched assertions; live controller not exercised.	**PASS**

ID	Test / expected result	Observed result	Status
TC-DUP-04	Duplicate attempt does not mutate the existing in-memory account fixture.	Fixture/password assertions passed; database not exercised.	**PASS**
TC-DUP-05	In-memory new-account fixture accepts matching bcrypt password and active status.	Simulation passed; registration/login API not exercised.	**PASS**

6. Defect Density Analysis

Defect density requires a confirmed defect count and an agreed software-size measure, such as KLOC, measured for the same scope. A verified project/module size was not collected for this run. The two typography findings are pending requirement-owner triage, so presenting a numeric defect-density value would be misleading.

Formula: Defect Density = confirmed defects ÷ software size

Component / module	Observed findings	Verified size metric	Density	Action
Typography / font selector	2 conformance findings pending owner confirmation; 1 exact-string test discrepancy	Not measured	Not calculated	Confirm acceptance criteria and measure module size only if density reporting is required.
Other tested modules	0 failed assertions in selected scripts	Not measured	Not calculated	Do not interpret passing helper/source assertions as exhaustive defect-free status.
Project total	3 grouped findings from 4 failed assertions	Not measured	Not calculated	Reconcile triage status and code-size scope before producing a density metric.

7. Priority–Severity Matrix

Severity describes impact. Priority describes urgency. This proposed matrix is a triage aid; it does not imply that a finding has been assigned by the project owner.

Priority \ Severity	Critical	High	Medium	Low
P1 — Urgent	Block release; immediate fix.	Fix before release when a key workflow is blocked.	Escalate only if broad impact or mandatory acceptance criterion is at risk.	Normally not P1; document exceptional reason.
P2 — High	Normally P1; document exception.	Fix in current cycle.	Fix before submission/demo if requirement is confirmed.	Schedule only when user impact warrants it.
P3 — Medium	Re-triage; normally unsuitable.	Re-triage or provide mitigation.	Plan in next suitable cycle.	Fix when capacity allows.
P4 — Low	Re-triage; normally unsuitable.	Re-triage; normally unsuitable.	Backlog only if impact is limited.	Minor issue; backlog.

> **Suggested triage:** QA-TYP-01 and QA-TYP-02 may be P2 if the checked-in approved-font requirement is still binding for the current release. QA-TYP-03 is P3 unless the fallback rule is explicitly prohibited. Confirm with the project owner.

8. Test Scripts Present but Not Run in This Cycle

The following existing scripts were not executed as part of this bounded run. Their existence is not evidence that their cases passed. The database-reseeding smoke test was excluded to avoid changing database contents; live SMTP/network and broader integration tests were excluded to avoid external side effects.

Existing project script	Reason not run in this cycle	Status
`backend/scripts/smokeTest.js`	Script documentation says it reseeds the database; its fallback can use the configured MongoDB URI.	**NOT RUN**
`backend/scripts/testEmailAuthFlow.js`	Connects to configured MongoDB and performs SMTP configuration/connection checks.	**NOT RUN**
`backend/scripts/testSmtpConnection.js`	Verifies SMTP and sends a test email; external side effect.	**NOT RUN**
`backend/scripts/testVerificationAndSecurity.js`	Starts a service and connects to configured MongoDB; database environment was not verified as isolated.	**NOT RUN**
`backend/scripts/testRealtime.js`	Starts an HTTP/socket integration test server; excluded from the selected local-only run.	**NOT RUN**
`backend/scripts/testEndToEndAuthFlow.js`	Starts an integration server; excluded from this bounded run.	**NOT RUN**
`backend/scripts/testAdminRecoveryAndGuide.js`	Starts an integration server; excluded from this bounded run.	**NOT RUN**
`backend/scripts/testEmailAndFeatureAudit.js`	Not part of the five selected scripts; not counted or reported as passed.	**NOT RUN**

The listed names are from the repository. No test cases were fabricated for these unexecuted scripts.

9. Defect Tracking Workflow

1. Confirm the font requirements against the currently approved project specification.
2. Record each confirmed product defect separately, with reproduction steps, expected/actual behavior, environment, severity, priority, owner and evidence.
3. Keep test-contract discrepancies separate from product defects until acceptance criteria are reconciled.
4. Use a consistent lifecycle: New/Open → Triaged → In Progress → Fixed/Resolved → Retested → Closed; reopen if retesting fails.
5. After a fix or test update, rerun the relevant existing typography tests and record the fresh output.
6. Run integration scripts only after confirming an isolated test database and safe external-service configuration.
7. Do not claim tracker tickets were created or findings closed unless the team records and verifies those actions.

10. Key Observations and Conclusion

- All 32 cases in the five selected existing scripts were executed: 28 passed and 4 failed.
- All four failures concern font/typography assertions: missing Helvetica is reported twice, plus one Times-stack exact-string mismatch and one Roboto restriction failure.
- The 28 passing assertions passed within the scope of their individual scripts; several are static-source checks or in-memory simulations, not end-to-end service tests.
- No live database reseeding, SMTP delivery, or production workflow was attempted.
- Defect density was not calculated because no verified size metric or final owner-triaged defect count is available.

This report records observed evidence, not an overall product certification. Resolve the font-policy question, triage the three grouped findings, and rerun only the relevant existing tests. The remaining project scripts can be run later against explicitly isolated infrastructure with their side effects understood.

Role	Name / signature	Date
Prepared by		
Reviewed by		
Approved by		

End of report.